# Try a tool with two people

Task 1.1 / Prepare and draw

Name _____ Team _____ Date _____

## 01 Set up the investigation

Pick a repeatable job. Draw the tool touching the material and label where the hand applies an input.

Observe one use of the tool. Record who used it, what it acted on, and one dimension or count you can check.

| User / job | Dimension or count | Where it helps | Where the user adjusts |
|---|---|---|---|
| User A | | | |
| User B | | | |

## 02 Plan it on paper

Sketch the tool as used. Label its input and the job the user wants done.

**My prediction, chosen feature or design limit**

# Try a tool with two people

Task 1.1 / Test, compare and explain

Name _____ Team _____ Date _____

## 03 Run the test and record what happened

Have your partner perform the job. Record a point where the tool helps and a point where the user has to adjust. Give a second user the same job. Compare the two observations and refine your statement of need.

| User | Observed action | Need before / after revision |
|---|---|---|
| A | | |
| B | | |

## 04 Use the reading

**Ch. 1 | Welcome to the Designed World | printed pp. 3-10**
Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

## 05 Individual explanation

Name the user, the job, and one piece of evidence that would justify a design change.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.

# Hold a phone blank upright

Task 1.2 / Prepare and draw

Name _____ Team _____ Date _____

## 01 Set up the investigation

Use a folder and up to 12 in of tape. Sketch two support shapes. Label the opening, contact points and measured thickness of the blank.

Measure the blank's width and thickness to the nearest 1/16 in. Mark its display side.

| Feature | Blank (in) | Holder A (in) | Holder B (in) |
|---|---|---|---|
| Width | | | |
| Thickness / clearance | | | |

## 02 Plan it on paper

Sketch two holders. Label the support, opening, and a critical dimension. Choose the more promising design.

**My prediction, chosen feature or design limit**

# Hold a phone blank upright

Task 1.2 / Test, compare and explain

Name _____ Team _____ Date _____

## 03 Run the test and record what happened

Fold and tape the first holder. Set it on the bench with the blank's face visible. Keep enough material for a revision. Time an unsupported hold up to 30 s. Remove and replace the blank one-handed three times. Change one feature and repeat both tests.

| Version | Hold (0-30 s) | One-hand cycles 1 / 2 / 3 | Total successes |
|---|---|---|---|
| First | | | |
| Revised | | | |

## 04 Use the reading

**Ch. 2 | Birth of a New Technology | printed pp. 12-14**

Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

## 05 Individual explanation

Which change would you keep? Give a dimension with units and compare the two versions' hold times or successful cycles.

Keep these sheets with the work. Save the individual explanation in the lesson's Explain step.

# Turn a holder into a reproducible pattern

Task 1.2–1.3 / Prepare and draw

Name _____ Team _____ Date _____

## 01 Set up the investigation

Use solid outlines for cuts and dashed lines for folds. Pair a front view and side view with a flat pattern. Label all critical dimensions and units.

Measure a flat pattern and the usable sheet. Mark cut and fold lines differently.

| Stock length × width | Pattern area | Area-only estimate | Whole patterns fitted |
|---|---|---|---|
| Sheet 1 | | | |
| Revised layout | | | |

## 02 Plan it on paper

Draw front and side views. Include width, height, and the opening. Mark which view supplies each measurement.

**My prediction, chosen feature or design limit**

# Turn a holder into a reproducible pattern

Task 1.2–1.3 / Test, compare and explain

Name _____ Team _____ Date _____

## 03 Run the test and record what happened

Fit copies of the pattern on one sheet without overlap. Count whole parts and compare with an area-only estimate. Give your views to a partner. Ask them to locate a fold and one hidden dimension. Correct any ambiguity.

| Feature | Partner's reading | Correction to view or dimension |
|---|---|---|
| Fold position | | |
| Hidden dimension | | |

## 04 Use the reading

**Ch. 3 | Designs That Take Flight | printed pp. 19-26**

**Ch. 4 | Beyond Words | printed pp. 27-34**

Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

## 05 Individual explanation

Show the number of whole parts per sheet. Explain one reason it can differ from the area estimate.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.

# Write an organizer brief

Task 1.4–1.5 / Prepare and draw

Name _____ Team _____ Date _____

## 01 Set up the investigation

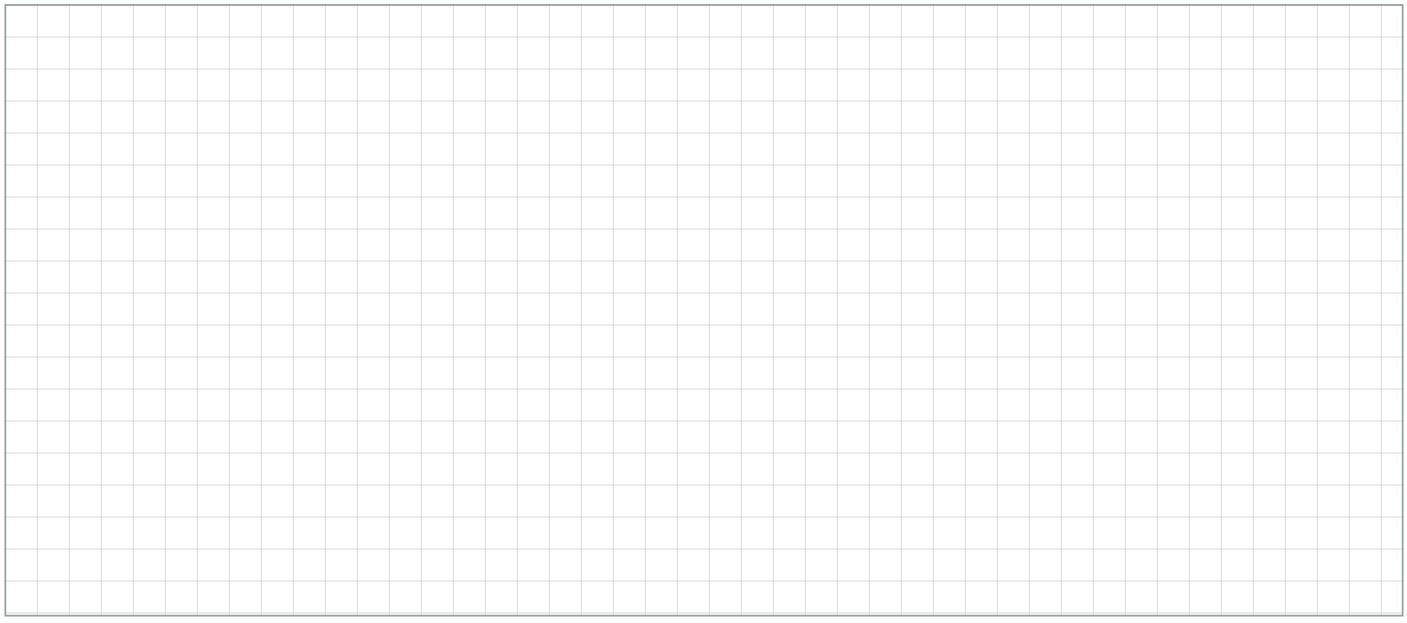
Arrange three objects on the available surface. Show the reach path. Agree on two signs of success and one limit with the actual user.

Measure three items. Ask the user which is hardest to retrieve. Record a quote or observed action.

| Item | Dimensions and units | User comment or observed action |
|---|---|---|
| Item 1 | | |
| Item 2 | | |
| Item 3 | | |

## 02 Plan it on paper

List the job, two measurable criteria, and a material or space limit. Sketch the area available.



**My prediction, chosen feature or design limit**

# Write an organizer brief

Task 1.4–1.5 / Test, compare and explain

Name _____ Team _____ Date _____

## 03 Run the test and record what happened

Compare an existing organizer with the brief. Have the user try its arrangement with the actual items. Read the brief back to the user. Replace a vague word such as easy with an action you can observe.

| Requirement | Observable test | User confirmation or change |
|---|---|---|
| Criterion 1 | | |
| Criterion 2 | | |
| Space / stock limit | | |

## 04 Use the reading

**Ch. 5 | The Art of Engineering | printed pp. 35-40**
**Ch. 6 | Bringing Designed Ideas to the Market | printed pp. 41-50**
Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

## 05 Individual explanation

Give one user requirement, its test, and the observation that led you to choose it.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.

# Compare two organizer layouts

Task 1.6–1.7 / Prepare and draw

Name _____ Team _____ Date _____

## 01 Set up the investigation

Draw layouts A and B at the same scale. Clear width equals outside width minus the thicknesses of the walls occupying that width. Test with the actual items.

Measure wall thickness and the outer footprint. Calculate the clear width left for storage.

| Layout | Outside width | Wall thicknesses | Clear width and units |
|---|---|---|---|
| A | | | |
| B | | | |

## 02 Plan it on paper

Each partner sketches a different layout. Compare both against the same criteria in the packet decision table.

**My prediction, chosen feature or design limit**

# Compare two organizer layouts

Task 1.6–1.7 / Test, compare and explain

Name _____ Team _____ Date _____

## 03 Run the test and record what happened

Mark the two layouts full size on paper. Place the items inside and trace the hand's access route. Test each layout for fit and retrieval. Use the results to choose a design and update its dimensions.

| Criterion | A result | B result | Which does the user prefer? |
|---|---|---|---|
| Fit | | | |
| Reach / retrieval | | | |
| Trade-off | | | |

## 04 Use the reading

**Ch. 7 | A Universe of Systems | printed pp. 51-60**
Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

## 05 Individual explanation

Which design won? Show one clearance calculation and name a trade-off the user accepts.

Keep these sheets with the work. Save the individual explanation in the lesson's Explain step.

# Make a cut list and inspect the fit

Task 1.8 / Prepare and draw

Name _____ Team _____ Date _____

## 01 Set up the investigation

Draw the joint and show a common reference edge for all cuts. Note the assembly order, the joint left accessible and any departure from the plan.

Transfer dimensions from one edge. Label parts and measure the actual thickness of one wall.

| Part / count | Planned L × W | Actual L × W | Units or adjustment |
|---|---|---|---|
| Base | | | |
| Wall 1 | | | |
| Wall 2 | | | |
| Divider / other | | | |

## 02 Plan it on paper

Create a cut list. Decide which part to fit first and which joint to leave accessible for inspection.

**My prediction, chosen feature or design limit**

# Make a cut list and inspect the fit

Task 1.8 / Test, compare and explain

Name _____ Team _____ Date _____

## 03 Run the test and record what happened

Cut and dry-fit the base and walls. Check the actual objects, then fasten the parts. Record one departure from the plan. Have a partner check a bay against its design dimension. Mark any interference or inaccessible item.

| Bay or part | Design size | Measured size | Fit with actual object |
|---|---|---|---|
| Bay 1 | | | |
| Bay 2 | | | |

## 04 Use the reading

**Ch. 8 | The Making of a New Balance Shoe | printed pp. 61-70**

Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

## 05 Individual explanation

Record a planned and an actual dimension. Explain whether the difference matters to this user.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.

# Watch a user retrieve an item

Task 1.9 / Prepare and draw

Name _____ Team _____ Date _____

## 01 Set up the investigation

Agree on a starting setup and success rule. Sketch the obstacle you suspect. Let the user try the task without coaching.

Choose a score for retrieval, such as successful attempts out of three. Record a timed baseline if time matters to the user.

| Attempt | Retrieved? | Time (s) | Observation / user words |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |
| Repeat | | | |

## 02 Plan it on paper

Agree on the starting arrangement and directions. Avoid coaching the tester during the trial.

**My prediction, chosen feature or design limit**

# Watch a user retrieve an item

Task 1.9 / Test, compare and explain

Name _____ Team _____ Date _____

## 03 Run the test and record what happened

Run the test with the actual user. Record what happened on each attempt and one comment about access. Repeat a trial to check that the problem recurs. Mark the feature you will change in the next version.

| Possible cause | How checked | What the result suggests |
|---|---|---|
| Feature to change | | |
| Alternative cause | | |

## 04 Use the reading

**Ch. 8 | The Making of a New Balance Shoe | printed pp. 61-70**
Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

## 05 Individual explanation

Use the trials to defend one revision. Name a second possible cause you checked.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.

# Hand over a revised organizer

Task 1.10–1.11 / Prepare and draw

Name _____ Team _____ Date _____

## 01 Set up the investigation

Mark the revision on a dimensioned drawing. List the units for material and price. Cost = amount used × price per unit.

Calculate material used or cost from a supplied classroom price. Label price assumptions.

| Material | Amount and units | Price per unit | Extended cost |
|---|---|---|---|
| Stock | | | |
| Other | | | |
| Total | | | |

## 02 Plan it on paper

Choose one revision from the test and mark its new dimension on the drawing.

**My prediction, chosen feature or design limit**

# Hand over a revised organizer

Task 1.10–1.11 / Test, compare and explain

Name _____ Team _____ Date _____

## 03 Run the test and record what happened

Modify the organizer. Have the same user repeat the original test. Record both versions' results. Exchange the final drawing and prototype. Ask a partner to reproduce a critical dimension and explain its purpose.

| Version | Retrieval test | Critical dimension | User or builder check |
|---|---|---|---|
| Original | | | |
| Revised | | | |

## 04 Use the reading

**Ch. 9 | Like Nature Intended | printed pp. 71-79**
Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

## 05 Individual explanation

Show one measured improvement or unresolved limit. Include the earlier result and the new one.

Keep these sheets with the work. Save the individual explanation in the lesson's Explain step.